

Main System Development Guideline

The Main System is the Pokedex Mobile App. It is the main application where all player progress is stored and displayed.

The Main System must include:

- Player name entry
- Player dashboard
- Starter Pokemon selection
- Pokedex / Pokémon log
- Pokemon level tracking
- Coin wallet
- Pokemon Store
- Inventory
- Evolution feature
- Facilitator manual logging
- Game history
- Result receiver from subsystems

Main System Assignment

Developer

Assigned Features

Developer 1 Player name entry, player dashboard, player records

Developer 2 Starter selection and Pokédex

Developer 3 Coins, Pokémon Store, inventory, and evolution

Developer 4 Facilitator logging, game history, and subsystem integration

Main System Step-by-Step Development

Step 1: Create the Start Page

The Start Page is the first screen that appears when the app is opened.

Required elements:

- [] App title
- [] Player name input field
- [] Start Journey button

Example layout:

Welcome to PokéJourney!

Enter Player Name:

[Start Journey]

Expected behavior:

1. Player enters name.
2. Player clicks Start Journey.
3. System checks if the name already exists.
4. If the name exists, load existing progress.
5. If the name does not exist, create new player record.
6. Redirect player to dashboard.

Step 2: Create Player Record

When a new player enters their name, the system must create a player record.

Player data should include:

player_id
player_name
section
assigned_journey
starter_pokemon_id
created_at

Minimum required fields:

player_id
player_name
created_at

Optional fields:

section
assigned_journey
starter_pokemon_id

Step 3: Create Player Dashboard

After entering a name, the player should be redirected to their dashboard.

The dashboard should display:

Player Name
Starter Pokémon
Total Pokémon Caught
Total Coins
Total Level Points
Buttons to other pages

Example:

Player: Jods
Starter: Charmander
Coins: 75
Pokémon Caught: 4
Total Levels: 35

[Pokédex]
[Starter Selection]
[Pokémon Store]
[Inventory]
[Game History]

Step 4: Create Starter Selection

The player must be able to choose or receive a starter Pokémon.

Starter Pokémon choices:

Region	Grass	Fire	Water
Kanto	Bulbasaur	Charmander	Squirtle
Unova	Snivy	Tepig	Oshawott
Paldea	Sprigatito	Fuecoco	Quaxly

Starter selection flow:

Player answers questionnaire

↓

System calculates result

↓

System recommends starter Pokémon

↓

Player confirms starter

↓

Starter is saved under player account

↓

Starter is added to player's Pokédex

Important rule:

A player should only receive one starter Pokémon.

If the player already has a starter, show:

You already have a starter Pokémon: Charmander

Step 5: Create Pokémon Database

The system must have a list of available Pokémon.

Pokémon data should include:

pokemon_id
pokemon_name
type
region
image
evolution_stage
evolves_to
required_stone

Example:

pokemon_id: 001
pokemon_name: Bulbasaur
type: Grass
region: Kanto
evolution_stage: 1
evolves_to: Ivysaur
required_stone: Leaf Stone

Step 6: Create Player Pokédex

The Pokédex shows all Pokémon owned by the player.

Each Pokémon record should display:

Pokémon Name

Type

Region

Level

Source

Status

Example:

Charmander

Type: Fire

Region: Kanto

Level: 15

Source: Starter

Status: Active

Possible Pokémon sources:

Starter

PokeReflex

PokeGuess

IRL Catch

Manual Log

Step 7: Create Level Tracking

Each player-owned Pokémon must have its own level.

Level rules:

Result	Level Gain
Correct / Win	+10 levels
Wrong / Partial	+1 level
Lose	+0 level

Important rules:

- [] Players cannot manually edit levels.
- [] Levels can only be updated through game results or facilitator logging.
- [] Every level update must be recorded in game history.

Step 8: Create Coin Wallet

Each player must have a coin balance.

Suggested coin rewards:

Action	Coins
Successfully catch Pokémon	+20 coins
Win mini game	+15 coins
Partial attempt	+5 coins
Lose or failed attempt	+0 coins

Coin rules:

- [] Coins increase when player wins or completes a challenge.
- [] Coins decrease when player buys from the Pokémon Store.
- [] Coin balance cannot go below zero.

Step 9: Create Pokémon Store

The Pokémon Store allows players to buy items using coins.

Suggested store items:

Item	Price	Purpose
Fire Stone	50 coins	Used for Fire-type evolution
Water Stone	50 coins	Used for Water-type evolution
Leaf Stone	50 coins	Used for Grass-type evolution
Thunder Stone	60 coins	Used for Electric-type evolution
Rare Candy	40 coins	Adds bonus level

Store purchase flow:

Player selects item

↓

System checks if player has enough coins

↓

If enough, deduct coins

↓

Add item to inventory

↓

Record purchase in game history

If coins are not enough, show:

Not enough coins.

Step 10: Create Inventory

The inventory shows all items owned by the player.

Example:

Fire Stone x1

Water Stone x2

Inventory rules:

- [] Purchased items must appear in inventory.
- [] Item quantity should increase when bought.
- [] Item quantity should decrease when used.

Step 11: Create Evolution Feature

The system should allow Pokémon to evolve.

Simple evolution rule:

A Pokémon can evolve if:

1. It reaches the required level.
2. The player owns the required evolution stone.

Suggested evolution requirements:

Evolution Stage	Requirement
First evolution	Level 20 + correct stone
Final evolution	Level 40 + correct stone

Type-based stone rule:

Pokémon Type	Required Stone
Fire	Fire Stone
Water	Water Stone
Grass	Leaf Stone

Electric

Thunder Stone

Evolution flow:

Player selects Pokémon



System checks level requirement



System checks required stone



If valid, Pokémon evolves



Stone quantity decreases



Evolution is recorded in game history

If not valid, show:

This Pokémon cannot evolve yet.

Step 12: Create Facilitator Manual Logging

The Facilitator Manual Logging feature is used for IRL games.

Facilitator should be able to log:

Player name
Game name
Result
Pokémon involved
Level gain
Coins earned
Notes

Manual logging examples:

Player: Jods
Game: IRL Catch a Pokémon
Result: Caught
Pokémon: Pikachu
Reward: +20 coins
Player: Jods
Game: Pokémon Showdown Battle
Result: Win
Pokémon: Charmander
Reward: +10 levels, +20 coins

Manual logging must update:

[] Player's Pokédex, if Pokémon was caught
[] Pokémon level, if level was earned
[] Coin wallet, if coins were earned
[] Game history

Step 13: Create Game History

Game history records every important action.

Examples of history logs:

Jods selected Charmander as starter.

Jods caught Squirtle through PokeReflex.

Jods gained +10 levels from Battle Predictor.

Jods earned +15 coins.

Jods bought Fire Stone.

Jods evolved Charmander.

Game history should include:

history_id
player_id
pokemon_id
game_name
result
level_gain
coins_earned
source_system
created_at

Step 14: Create Result Receiver for Subsystems

The Main System must receive results from Subsystem 1 and Subsystem 2.

Recommended result format:

```
{
  "playerName": "Jods",
  "pokemonName": "Squirtle",
  "gameName": "PokeReflex",
  "result": "caught",
  "levelGain": 0,
  "coinsEarned": 20,
  "sourceSystem": "catching_subsystem"
}
```

For leveling games:

```
{
  "playerName": "Jods",
  "pokemonName": "Charmander",
  "gameName": "Battle Predictor",
  "result": "correct",
  "levelGain": 10,
  "coinsEarned": 15,
  "sourceSystem": "leveling_subsystem"
}
```

The Main System must:

- [] Find the player by playerName.
- [] Update Pokémon record if needed.
- [] Update level if needed.
- [] Update coins if needed.
- [] Save the result in game history.